

SUOLOVUOPMI LULIT, Norway

WMO: 010580

Lat: 69.579N

Lon: 23.534E

Elev: 381

StdP: 96.83

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.9	-30.1	-36.3	0.1	-32.7	-33.2	0.2	-29.8	13.2	-5.9	11.9	-5.7	1.7	130	0.820

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	21.6	13.7	19.4	13.0	17.2	12.1	15.2	19.3	13.8	17.5	12.7	16.4	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	10.0	16.9	12.0	9.1	15.1	10.8	8.5	14.4	43.7	19.2	39.9	17.6	37.0	16.5	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.0	7.8	-36.0	25.4	2.8	2.9	-38.1	27.5	-39.7	29.2	-41.3	30.9	-43.4	33.0
			-35.9	16.6	2.9	1.8	-38.0	17.9	-39.6	18.9	-41.2	19.9	-43.3	21.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-13.7	-13.2	-10.0	-4.3	2.4	7.7	11.6	9.8	5.7	-1.6	-8.0	-9.9
	DBStd	10.31	7.60	7.80	6.64	4.96	4.05	3.73	3.61	3.34	3.36	4.78	5.77	6.64
	HDD10.0	4478	735	648	620	429	238	91	24	46	132	358	539	617
	HDD18.3	7383	993	882	878	679	494	319	212	266	378	617	789	876
	CDD10.0	139	0	0	0	0	3	21	72	39	4	0	0	0
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	39	0	0	0	0	2	1	28	7	0	0	0	0
	CDH26.7	6	0	0	0	0	0	0	4	1	0	0	0	0
Wind	WSAvg	3.5	3.6	3.8	4.0	3.7	3.7	3.7	3.4	3.0	3.4	3.2	3.2	3.6
Precipitation	PrecAvg	458	30	26	28	23	31	51	66	58	39	44	32	34
	PrecMax	546	62	44	50	49	60	97	148	119	65	98	60	70
	PrecMin	315	11	3	12	6	4	15	18	29	7	13	7	16
	PrecStd	56	12	10	8	10	15	22	34	23	15	20	15	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	3.3	3.9	7.8	20.1	21.6	26.9	23.4	16.6	9.3	3.2	3.8
		MCWB	0.7	1.7	1.0	4.2	11.2	12.2	17.4	15.8	11.2	6.9	2.1	2.0
	2%	DB	-0.2	0.8	1.5	5.2	15.0	19.0	22.7	20.1	14.2	7.5	2.1	1.8
		MCWB	-1.2	-0.7	-0.2	2.1	8.1	11.9	14.8	13.8	10.4	5.8	1.2	0.4
	5%	DB	-2.0	-1.0	0.1	3.7	11.1	16.3	20.3	17.6	12.5	5.8	0.6	0.0
		MCWB	-2.9	-1.9	-1.3	1.2	6.3	10.6	13.7	12.7	9.6	4.4	-0.2	-0.9
	10%	DB	-3.8	-2.8	-1.2	2.4	8.5	14.2	18.1	15.6	11.1	4.3	-0.8	-1.3
		MCWB	-4.4	-3.7	-2.4	0.3	4.8	9.9	12.9	12.1	8.6	3.3	-1.4	-2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.9	1.7	1.2	4.5	11.7	14.1	18.3	17.2	12.6	6.9	2.5	2.2
		MCDB	2.0	3.2	3.3	7.9	20.2	17.9	25.2	23.1	14.9	8.6	3.0	3.7
	2%	WB	-1.2	-0.6	0.0	2.4	8.6	13.0	16.1	14.4	11.1	5.7	1.2	0.6
		MCDB	-0.2	0.6	1.5	4.8	14.5	16.4	20.9	17.5	13.4	7.2	2.0	1.6
	5%	WB	-2.8	-1.9	-1.2	1.4	6.5	11.6	14.5	13.2	9.9	4.5	-0.1	-0.9
		MCDB	-2.1	-1.1	0.0	3.5	10.6	16.0	18.8	16.5	12.1	5.6	0.5	0.0
	10%	WB	-4.4	-3.6	-2.3	0.5	5.1	10.0	13.3	12.2	8.7	3.3	-1.3	-2.1
		MCDB	-3.9	-2.9	-1.2	2.2	8.3	13.5	17.4	15.0	10.7	4.3	-0.7	-1.4
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.1	10.5	10.0	7.7	7.6	8.5	8.4	6.7	5.5	7.6	8.6
		MCDBR	8.9	9.1	7.8	8.8	12.6	11.2	12.6	12.3	9.2	5.9	6.0	7.4
		MCWBR	8.4	8.3	6.6	6.3	7.0	5.6	5.7	6.7	5.6	4.6	5.7	6.7
	5% WB	MCDBR	8.8	8.7	7.7	8.5	11.9	10.1	11.1	10.2	8.0	5.4	5.9	7.2
		MCWBR	8.4	8.0	6.7	6.1	6.9	5.4	5.6	6.0	5.3	4.4	5.6	6.7
Clear-Sky Solar Irradiance	taub	0.181	0.229	0.277	0.301	0.319	0.323	0.330	0.316	0.288	0.241	N/A	N/A	
	taud	1.911	2.068	2.171	2.223	2.361	2.500	2.505	2.524	2.558	2.227	N/A	N/A	
	Ebn at Noon	27	517	734	834	866	876	854	827	759	505	0	0	
	Edh at Noon	3	41	75	98	97	88	85	74	54	34	0	0	
All-Sky Solar Radiation	RadAvg	0.01	0.37	1.79	3.78	5.04	4.88	4.47	3.12	1.63	0.49	0.00	0.00	
	RadStd	0.00	0.03	0.09	0.18	0.45	0.40	0.43	0.25	0.14	0.04	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (29 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page